Computer Graphics (CSC214)

Unit - 09: Introduction to Virtual Reality

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Credit Hours: 3 hrs

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Introduction to Virtual Reality

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Introduction to VR:

Introduction to VR:

Virtual reality is a system for providing an **interactive exploration** of a three dimensional **(3-D)** virtual environment.

- Virtual reality is an environment that is **simulated** by a computer, trying to **imitate** the real thing.
- Virtual reality refers to a high-end user interface that involves **real-time simulation** and **interactions** through multiple sensorial channels

A typical VR system consists of **six main components** grouped into two:

1. **Internal Components:**
   1. Virtual World
   2. Graphics Engine
   3. Simulation Engine
   4. User Interface

2. External Components:

I. User Inputs
II. User Outputs

VR Components:

- **Virtual world:**
  - A scene database containing the **geometric representations and attributes** for all objects within the environment.

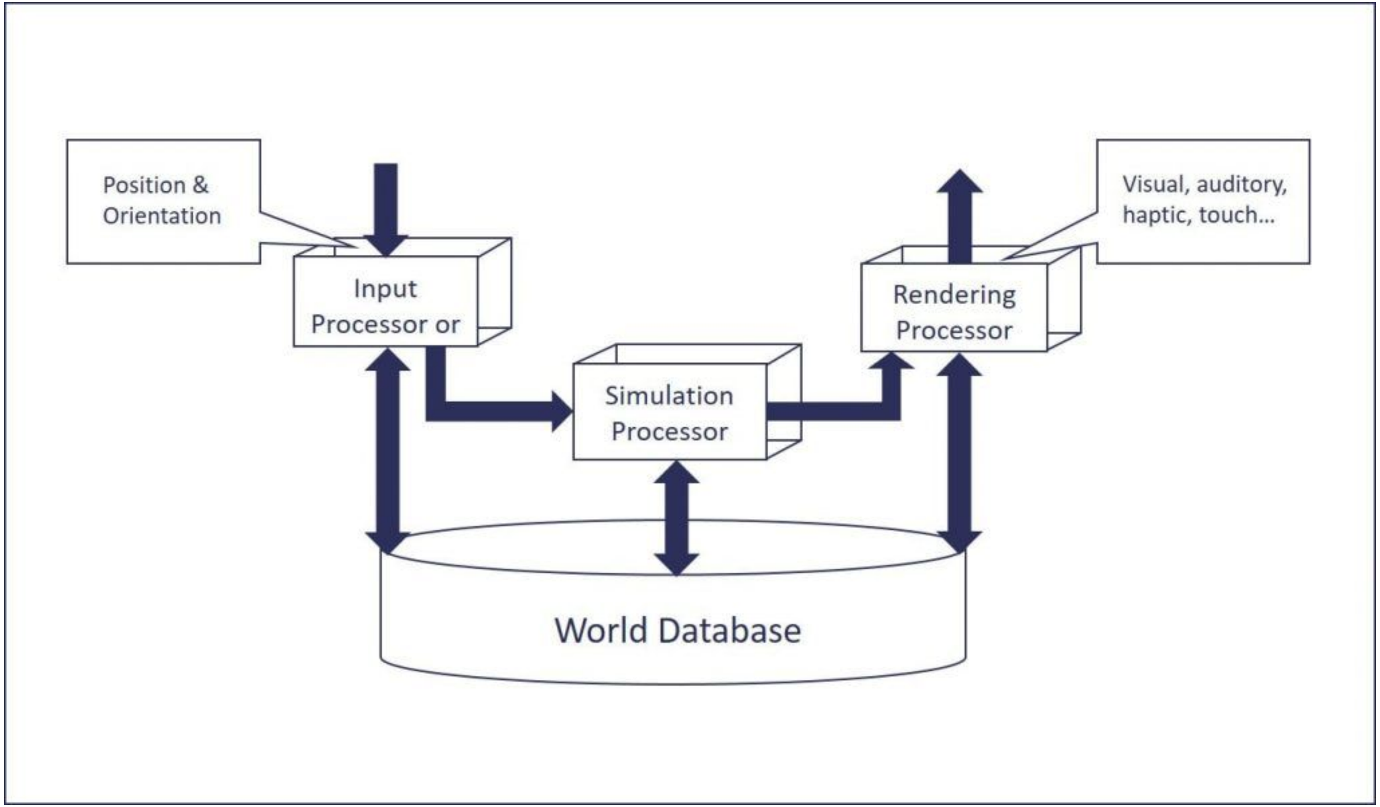
- **Graphics Engine:**
  - Responsible for actually **generating the image** or scene, which a **viewer** will see.
  - Usually the **scene database** and the **viewer's current position** and **orientation** is taken into account.
  - It also includes **other information** from the scene database e.g. sound, special effects textures etc.

VR Components:

- **Simulation Engine:**
    - Does **most of the work** required to maintain virtual environment.
    - Concerned purely with the dynamics of the environment
        - How it changes over time?
        - How it responds to the user's actions?
    - This includes handling interactions, physical simulations (gravity, inertia).

- **User interface:**
    - Controls how the user **navigates and interacts** with this virtual environment.

VR Architecture:



VR Headset Components:

1. Sensors
   a. Magnetometer
   b. Accelerometers
   c. Gyroscopes
2. Lenses
3. Display Screens
4. Processing
   a. Input Processor
   b. Simulation Processor
   c. Rendering Processor

For Detail Explanation Read KEC Book Page Number 228.

Types of VR System:

- **Desktop VR (Non-Immersive):**
  - Based on the concept that the potential user **interacts with** the **computer** screen **without** being fully immersed and surrounded by the computer generated environment.
  - **Applications domains** involve architecture, industrial design, data visualization
  - Less cost and involves less use of interacting technology.

- **Immersive VR:**
  - Completely **immerse** the user's personal viewpoint inside the **virtual 3D** world.
  - The user has **no visual contact** with the physical world.
  - Often equipped with a Head Mounted Display (**HMD**).

Types of VR System:

- **Telepresence:**
    - A variation of visualizing **complete computer generated** worlds.
    - Links **remote sensors in the real world** with the senses of a human operator.
    - The remote sensors might be located on a robot. Useful for performing operations in **dangerous environments**.

- **Distributed VR:**
    - A simulated world runs on several computers which are **connected over network** and the people are **able to interact** in real time, sharing the same virtual world.

Types of VR System:

- **Augmented Reality (Mixed reality):**
  - The **seamless merging** of real space and virtual space.
  - Integrate the computer-generated virtual objects **into the physical world** which become in a sense an equal part of our natural environment.
  - **Example:** Head Up Displays **(HUD)**
    - Used in modern military aircraft
    - These **superimpose flight data** such as altitude, air speed upon the pilots field of view.
    - This can be on a cockpit mounted display or upon the **pilot's helmet visor.**

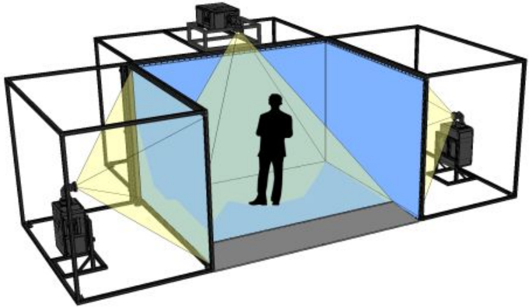
Technologies of VR – Hardware :

- **Head-Mounted Display (HMD):**
  - A **Helmet** or a **face mask** providing the visual and auditory displays.
  - Use LCD or CRT to display stereo images.
  - May include **built-in head-tracker** and stereo headphones.

- **Binocular Omni-Orientation Monitor (BOOM):**
  - **Head-coupled** stereoscopic display device.
  - Uses CRT to provide high-resolution display.
  - Conventional to use.
  - Fast and accurate built-in tracking.

Technologies of VR – Hardware :

- **Cave Automatic Virtual Environment (CAVE):**
    - Provides the illusion of immersion by projecting stereo images on the **walls and floor** of a **room-sized cube**.
    - A **head tracking system** continuously adjust the stereo projection to the **current position** of the leading viewer.



- **Data Glove:**
    - Outfitted with **sensors on the fingers** as well as an overall position/orientation tracking equipment.
    - Enables **natural interaction** with virtual objects by hand gesture recognition.

Technologies of VR – **Hardware** :

- **Control devices:**
  - Control virtual objects in **3 dimensions**.

▶ Technologies of VR – **Software** :

- **Toolkits:**
    - Programming libraries.
    - Provide function libraries (C & C++)

- **Authoring system:**
    - Complete programs with graphical interfaces for creating **worlds without resorting** to detailed programming.

Technologies of VR – Software :

- **Virtual Reality Modeling Language (VRML):**
  - **Standard language** for interactive simulation within the World Wide Web.
  - Allows to create "virtual words" networked via the internet and hyperlinked with the **World Wide Web**.
  - Aspects of virtual world display, **interaction and internetworking** can be specified using VRML without being dependent on special gear like HMD.

Applications of VR:

- **Flight Simulation:**
  - For pilot training
  - Safe and realistic
  - Risk free
- **Medicine:**
  - Practice performing surgery.
  - Perform surgery on a remote patient.
  - Teach new skills in a safe, controlled environment.
- **Education and Training:**
  - Driving simulators.
  - Ship simulator
  - Flight simulator

Applications of VR:

- **Engineering and Design:**
  - CAD and CAM
  - View products as it would be seen when manufactured.
- **Human Factor Modeling:**
  - Used to model human behavior in the design of new products or buildings
  - E.g. simulation of fire in a building and a user can view how the virtual occupants react to the emergency
- **Visualization:**
  - Data Visualization
- **Military**

Advantages of VR:

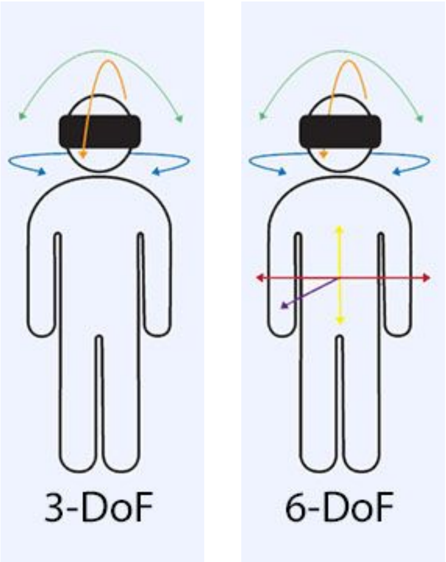
- **Enhanced Learning and Training:** VR provides immersive and interactive environments for education and training, making complex subjects easier to understand and practice without real-world risks.
- **Improved Entertainment Experiences:** VR offers highly immersive experiences in gaming and movies, allowing users to feel as if they are part of the story or action.
- **Innovative Design and Visualization:** Architects, engineers, and designers can use VR to visualize their creations in a 3D space, enabling better design decisions and client presentations.
- **Therapeutic Applications:** VR is used in healthcare for pain management, physical therapy, and treating mental health conditions such as PTSD and anxiety through controlled, immersive environments.
- **Enhanced Remote Work and Collaboration:** VR enables virtual meetings and collaboration, making remote work more interactive and engaging compared to traditional video conferencing.

Disadvantages of VR:

- **High Cost:** VR equipment, including headsets and powerful computers, can be very expensive, making it less accessible to the general public.
- **Health Issues:** Prolonged use of VR can cause physical discomforts such as eye strain, headaches, and nausea (motion sickness), affecting user health.
- **Limited Mobility:** Users are often confined to a small area to prevent collisions with real-world objects, which limits movement and the immersive experience.
- **Technical Limitations:** Despite advancements, VR still faces issues with resolution, latency, and field of view, and creating high-quality VR content is complex and resource-intensive.
- **Social Isolation:** VR experiences tend to be solitary, which can lead to social isolation as users spend less time interacting with others in the real world.

3D Positional Tracking:

- Positional tracking is a technology that allows a **device to estimate its position relative** to the environment around it.
- It uses a **combination of hardware and software** to achieve the detection of its absolute position.
- It is an essential technology for virtual reality (VR), making it possible to track movement with **six degrees of freedom (6DOF)**.
- Positional tracking VR technology brings various benefits to the VR experience. It can **change the viewpoint of the user** to reflect different actions like jumping, ducking, or leaning forward.
- It increases the connection between the physical and virtual worlds.



Visual Computation in Virtual Reality:

- Visual computation allows **interaction and control** by manipulating visual images, either as direct work objects or as objects representing other objects that are **not necessarily visual themselves.** These visual images can be photographs, 3D scenes, block diagrams, or simple icons.
- Visual computation generally describes:
    - **Computer environments** in which a visual paradigm, rather than a text paradigm, is used.
    - **Applications** dealing with large or numerous image files, such as **video sequences** and 3D scenes.

Virtual Reality vs Augmented Reality:

Virtual Reality(VR)	Augmented Reality(AR)
Virtual Reality creates a fully immersive digital environment or experience that simulates the real world or imaginary world.	Augmented Reality overlays digital information into the real world.
It generally requires a headset or a similar kind of device to immerse the user into the digital world.	It can be accomplished through smartphones or tablets with the help of AR apps.
The user is isolated from the real world while in VR.	The user is aware of the real world while experiencing AR.
It requires powerful hardware and software to create a realistic experience.	It requires relatively simple technology for the creation.
Examples: PlayStation VR, Samsung Gear VR, and HTC Vive.	**Examples:** Pokemon GO, Google Maps AR, and IKEA App.

I^3 in Virtual Reality

I^3 in Virtual Reality refers to the **three core components** that define a VR system's effectiveness and user experience

- **Immersion**
  - The sensation of **being physically present** in a non-physical world.
  - Achieved through visual, auditory, and sometimes haptic (touch-based) feedback.
  - The more immersive the experience, the more the user feels "inside" the virtual world.
- **Interaction**
  - The ability of the **user to engage** with the virtual environment.
  - Involves input devices (controllers, gloves, motion sensors) that let users manipulate virtual objects, navigate environments, or trigger actions.
- **Imagination**
  - The **mental and creative engagement** of the user.
  - The richness of the virtual environment should stimulate curiosity, problem-solving, and emotional involvement, enhancing the illusion of reality.

Basic Principles of Animation & Types of Animation

9.5

Animation:

Animation is the process of displaying **still images** in a **rapid sequence** to create the **illusion** of movement.

- Animation is especially useful for illustrating concepts that **involve movement**.
  Example: Concept such as playing guitar or hitting a cricket ball is difficult to explain using a text or a single photograph. Animation makes it easier to portray these aspects of multimedia application.
- A person who creates animations is called **animator**. He/She use various computer technologies to capture the pictures and then to animate these in the desired sequence.

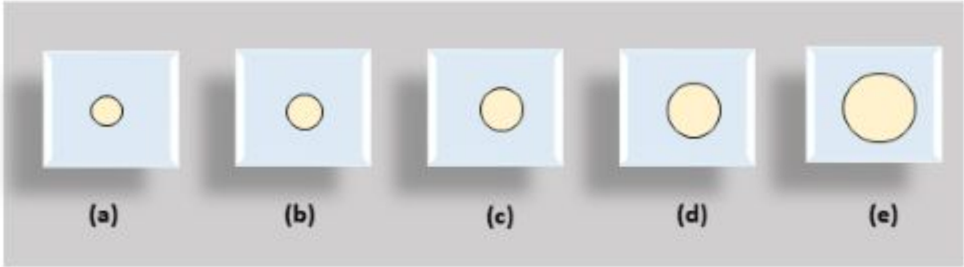


Fig: Change in Size

Steps of Animation Sequence Designing:

1. Animation Story Layout:
2. Animation Object Definition
3. Frame Specification
4. Generation of In-Between Frames

Steps of Animation Sequence Designing:

1. **Animation Story Layout:**

   Description:

   - The animation story layout, or storyboard, **outlines** the motion sequence of objects as a series of basic events.
   - **For instance**, in a cricket-playing scene, it defines the actions and motions of batting, bowling, fielding, running, etc.
   - The storyboard may include models, sketches, or verbal descriptions.

2. **Animation Object Definition:**

   Description:

   - After preparing the storyboard, all objects used in the animation scene are defined in detail. This includes each object's **dimensions, shapes, colors, movements**, and any additional necessary information.
   - For a cricket scene, this involves defining the players' dimensions, the colors of their uniforms, and the dimensions of the ball, bat, stumps, etc.

Steps of Animation Sequence Designing:

3. Frame Specification:

Description:

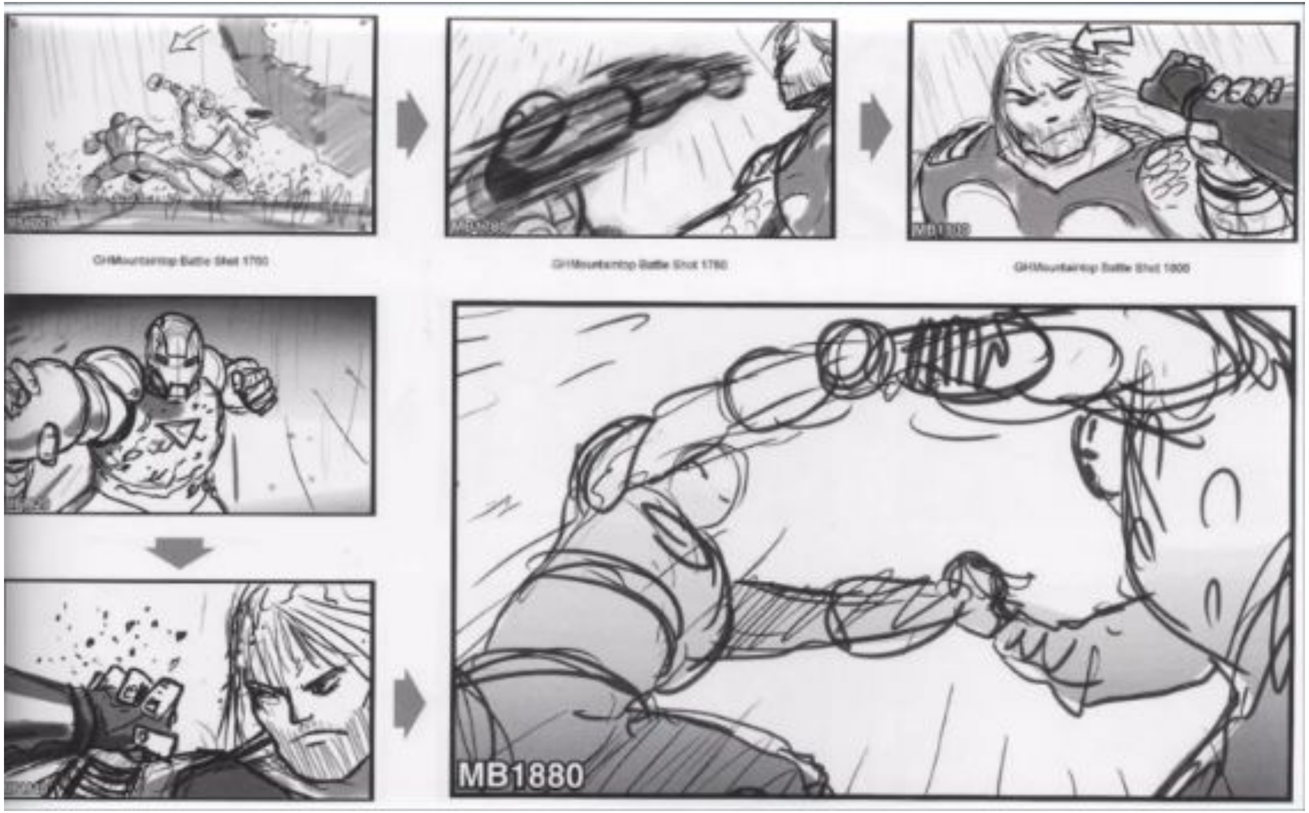
- Key frames are specified and created in detail, showing the **position, color, shapes,** etc., of all objects at particular points in the animation.
- These frames serve as **important reference points** in the animation sequence

4. Generation of In-Between Frames:

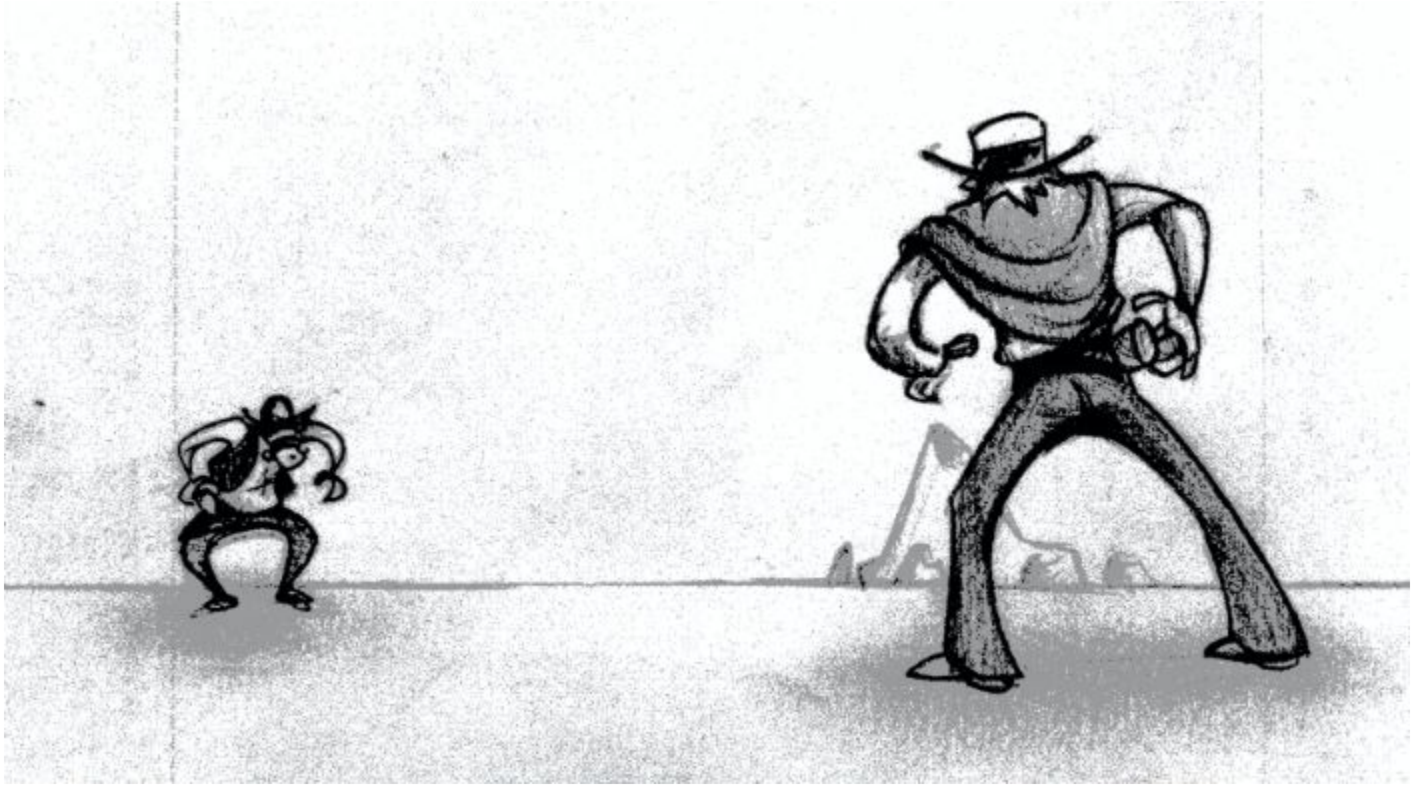
Description:

- In-between frames are created to **fill the gaps** between key frames, often using geometric **transformations.**
- Approximately 1500 frames are needed for one minute of film.

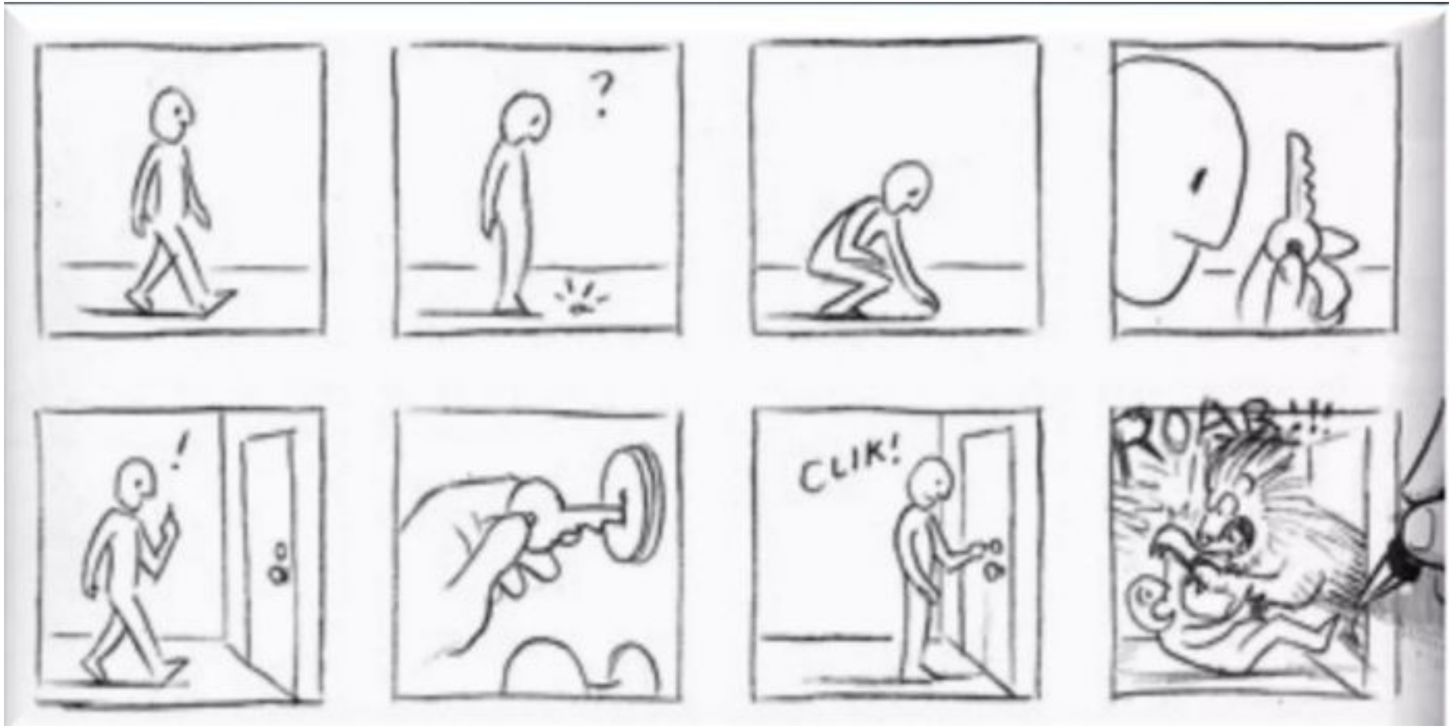
StoryBoard Layout



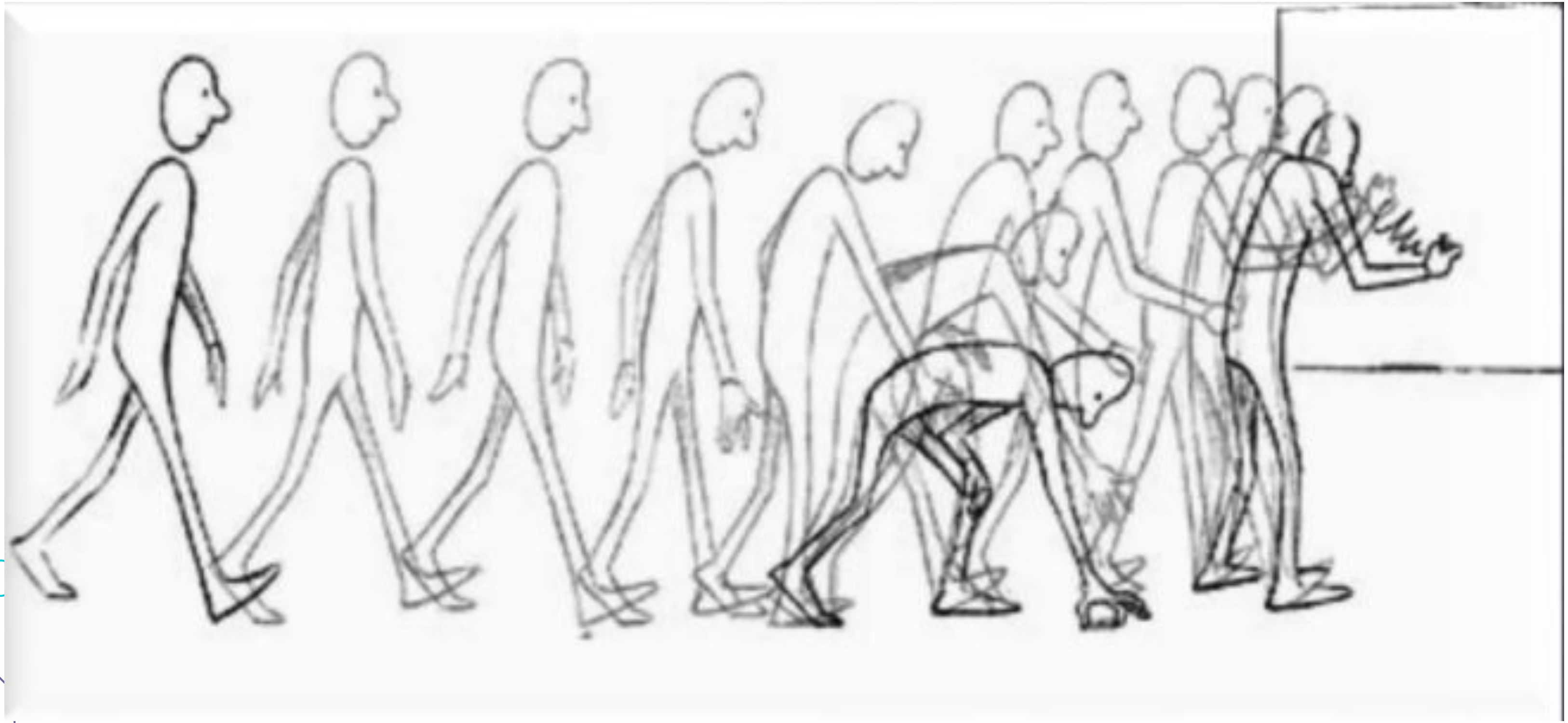
Object Definition:



Key Frames



In-Between Frames:



PYQs

1. What is animation? Explain the steps to create the animation sequence. 2018 (5 Marks)
2. What is a VR? Explain types of VR. (2+3) 2021
3. Explain the architecture of VR system with necessary components. 2020 (5)

Assignment - 9:

1. Differentiate between virtual reality and augmented reality with examples. 2081
2. What is animation? Explain the steps to create the animation sequence. 2018 (5 Marks)
3. What is a VR? Explain types of VR. (2+3) 2021
4. Explain the architecture of VR system with necessary components. What are Pros and Cons of VR? 2020 (5)
5. Describe various hardware and software components of VR.
6. List and explain application areas of VR. How can we apply VR for education?